

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK

EXTENSION 2

MATHEMATICS

MARCH 2006

Instructions

- Attempt all questions
- Answers to be written on the paper provided
- Start each question on a new page
- Marks may not be awarded for careless or badly arranged working
- Indicated marks are a guide and may be changed slightly if necessary
- These questions must be handed in attached to the top of your solutions.

Q1	Q2	Q3	Total
/16	/17	/17	

Question 1 (16 marks)

a) If $z = -1 + i$ find, (4)

(i) $\bar{z}$

(ii) $|z|$

(iii) $\arg(iz)$

b) i. Express $z = -\sqrt{3} - i$ in modulus – argument form (3)

ii. Hence write z^{12} in the form $x + iy$, where x and y are real

c) Find all complex numbers z , such that $z^3 = 64i$ (3)

- d) Given that a and b are real numbers, find a and b if, $\frac{5+2i}{a+bi} = 1+i$ (3)
- e) On an argand diagram shade the region containing all points representing complex numbers z such that $2 \leq \operatorname{Re}(z) \leq 5$ and $-2 \leq \operatorname{Im}(z) \leq 4$

Question 2 (17 marks) (Start a new page)

- a) Evaluate $\int_0^1 x\sqrt{1-x} \, dx$ using a suitable substitution. (3)

- b) Consider the ellipse $3x^2 + 4y^2 = 12$ (6)

- Determine the eccentricity of the ellipse
- Find the coordinates of the foci S and S' and also the equation of the directrices.
- Sketch the ellipse showing all important information.

c)

- Show that the equation of the tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at the point

$P(3\cos\theta, 2\sin\theta)$ is given by $\frac{x\cos\theta}{3} + \frac{y\sin\theta}{2} = 1$ (3)

The ellipse above cuts the y axis at the points A and B (5)

The tangents to the ellipse at A and B meet the tangent to the ellipse at $P(3\cos\theta, 2\sin\theta)$ at the points C and D respectively.

- Draw a neat diagram showing the positions of P, A, B, C and D .
- Show that $AC \times BD = 9$

Start a new page

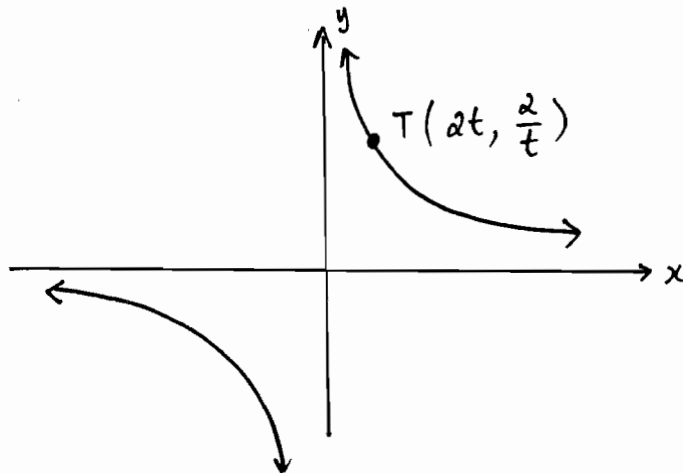
Question 3 (17 marks)

- a) Show that the locus specified by $3|z - (4 + 4i)| = |z - (12 + 12i)|$ is a circle, (4)

Write down its radius and the coordinates of its centre

- b) Find the equation of the tangent to the curve $x^2 + xy^2 - 6y = 0$ at the point (2,1) (4)

- c) Consider the diagram below



- i. Show that the tangent to the hyperbola $xy = 4$ at the point $T(2t, \frac{2}{t})$ has equation $x + t^2y = 4t$. (2)
- ii. This tangent cuts the x -axis at point Q. Show that the line through Q which is perpendicular to the tangent at T has equation $t^2x - y = 4t^3$. (2)
- iii. This line through Q cuts the rectangular hyperbola at the points R and S. Show that the midpoint M of RS has coordinates $M(2t, -2t^3)$. (2)
- iv. Find the equation of the locus of M as T moves on the rectangular hyperbola, stating any restrictions that may apply. (3)

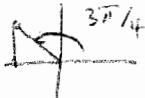
End of Test

Solutions

Question 1

a) $\bar{z} = -1 - i$ ✓

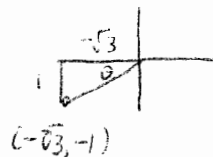
$\sqrt{2}$ ✓

$\arg(z) = \frac{3\pi}{4}$ 

$$\begin{aligned}\arg(iz) &= \pi/2 + 3\pi/4 \\ &= 5\pi/4 \quad \checkmark \\ &= -\frac{3\pi}{4} \quad \checkmark\end{aligned}$$

b) $z = -\sqrt{3} - i$

$\angle = -\frac{5\pi}{6}$



$|z| = 2$

$\therefore z = 2 \operatorname{cis} \left(-\frac{5\pi}{6} \right)$ ✓

ii. $z^{12} = 2^{12} \operatorname{cis} \left(12 \times -\frac{5\pi}{6} \right)$ ✓

$$\begin{aligned}&= 4096 (1 + i0) \\ &= 4096 \quad \checkmark \text{ ie } 2^{12}\end{aligned}$$

c) $z^3 = 64i$

$z^3 - 64i = 0$

$z^3 + (4i)^3 = 0$

$(z + 4i)(z^2 - 4iz - 16) = 0$

$z = -4i \checkmark, z = 2i \pm 2\sqrt{3}$ ✓

d) $5 + 2i = (1+i)(a+bi)$ ✓

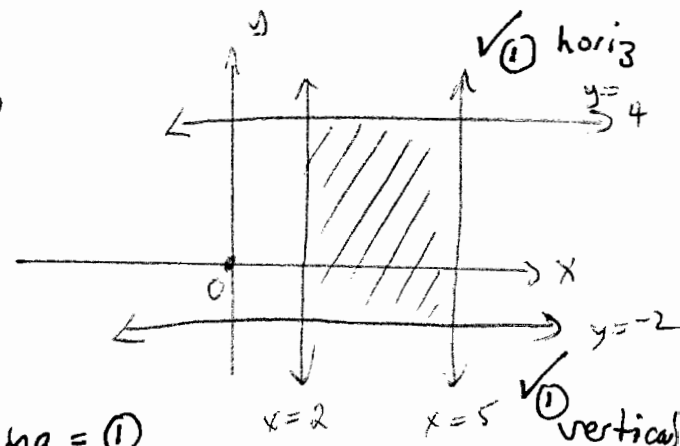
$= a + bi + ai - b$

$\therefore a - b = 5$ ✓

$a + b = 2$

$2a = 7$

e)



shading = ①

Question 2

a) $u = 1 - x$

$x = 1 - u$

$\frac{du}{dx} = -1$

$x = 1 \quad u = 0$

$x = 0 \quad u = 1$

$-du = dx$

①

$\therefore \int_1^0 (1-u)\sqrt{u} \cdot -du$ ①

$= \int_1^0 \sqrt{u} - u^{3/2} du$

$= \left[\frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^1$

$= \left(\frac{2}{3} - \frac{2}{5} \right) - (0)$

$= \frac{4}{15}$

①

$$b) \frac{x^2}{4} + \frac{y^2}{3} = 1 \quad \checkmark$$

$$a = 2 \quad b = \sqrt{3}$$

$$i. \quad b^2 = a^2(1 - e^2)$$

$$\frac{3}{4} = 1 - e^2$$

$$e^2 = \frac{1}{4} \quad \checkmark$$

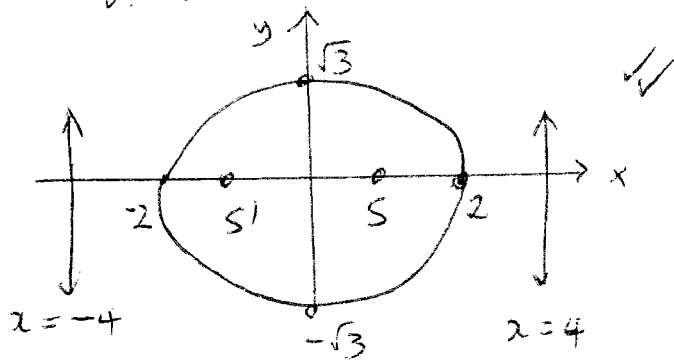
$$e = \frac{1}{2}$$

$$S = (ae, 0) \quad S'(-ae, 0)$$

$$S = (1, 0) \quad S'(-1, 0) \quad \checkmark$$

$$\text{directrices } x = \pm a/e$$

$$\therefore x = \pm 4 \quad \checkmark$$



$$c) \frac{2x}{9} + \frac{2y}{4} \cdot \frac{dy}{dx} = 0 \quad \checkmark$$

$$\frac{dy}{dx} = -\frac{2x}{9} \times \frac{4}{2y}$$

$$= -\frac{4}{9} \frac{x}{y}$$

$$M_T = \frac{-4 \cdot 3 \cos \theta}{3 \cdot 9 \cdot 2 \sin \theta}$$

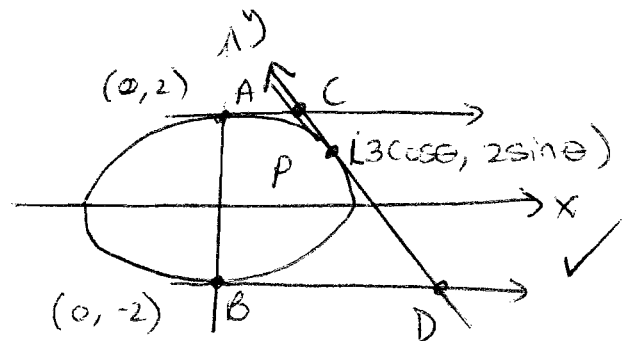
$$= \frac{-2 \cos \theta}{3 \sin \theta} \quad \checkmark$$

$$y - 2 \sin \theta = \frac{-2 \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$$

$$3 \sin \theta y - 6 \sin^2 \theta = -2 \cos^2 \theta x + 6 \cos^2 \theta$$

$$2 \cos^2 \theta x + 3 \sin^2 \theta y = 6$$

$$\therefore \frac{\cos^2 \theta x}{3} + \frac{\sin^2 \theta y}{2} = 1$$



$$\text{tangent at A } y = 2$$

$$\text{at B } y = -2$$

$$\text{tangent at P}$$

$$\frac{x \cos^2 \theta}{3} + \frac{y \sin^2 \theta}{2} = 1 \quad (1)$$

$$\text{Finding C sub } y = 2 \text{ into (1)}$$

$$\frac{x \cos^2 \theta}{3} + \sin^2 \theta = 1$$

$$\frac{x \cos^2 \theta}{3} = 1 - \sin^2 \theta$$

$$\checkmark \quad x = \frac{3(1 - \sin^2 \theta)}{\cos^2 \theta}$$

$$C = \left[\frac{3(1 - \sin^2 \theta)}{\cos^2 \theta}, 2 \right]$$

$$\text{Finding D sub } y = -2 \text{ into (1)}$$

$$\checkmark \quad \frac{x \cos^2 \theta}{3} - \sin^2 \theta = 1$$

$$D = \left[\frac{3(1 + \sin^2 \theta)}{\cos^2 \theta}, -2 \right]$$

$$\begin{aligned}
 KC \times BD &= \frac{3(1-\sin\theta)}{\cos\theta} \times \frac{3(1+\sin\theta)}{\cos\theta} \\
 &= 9 \frac{(1-\sin^2\theta)}{\cos^2\theta} \\
 &= 9
 \end{aligned}$$

Question 3

$$\begin{aligned}
 i) \quad 3|x+iy-4-4i| &= |x+iy-12-12i| \\
 3|(x-4) + (y-4)i| &= |(x-12) + (y-12)i| \\
 9[(x-4)^2 + (y-4)^2] &= (x-12)^2 + (y-12)^2
 \end{aligned}$$

$$\begin{aligned}
 9[x^2-8x+16+y^2-8y+16] &= x^2-24x+144 \\
 &\quad + y^2-24y+144
 \end{aligned}$$

$$8x^2 - 48x + 8y^2 - 48y = 0$$

$$x^2 - 6x + y^2 - 6y = 0$$

$$(x-3)^2 + (y-3)^2 = 18$$

$$\therefore \text{centre } (3,3) \quad \text{radius} = 3\sqrt{2}.$$

b) Implicit diff

$$2x + 1 \cdot y^2 + x \cdot 2y \cdot \frac{dy}{dx} - 6 \frac{dy}{dx} = 0 \quad (1)$$

$$2x + y^2 + \frac{dy}{dx}(2xy-6) = 0$$

$$\frac{dy}{dx} = \frac{-2x-y^2}{2xy-6} \quad (1)$$

$$\therefore \text{at } (2,1) \quad M_T = \frac{-2(2)-1}{2(2)(1)-6}$$

$$= \frac{5}{2} \quad (1)$$

$\therefore$ eq tangent

$$u-1 = 5/2(x-2) \quad \leftarrow (1)$$

$$c) \quad y = 4/x \quad y' = -\frac{4}{x^2} \quad \text{at } x=2t$$

$$\begin{aligned}
 i. \quad M_T &= \frac{-4}{4t^2} \\
 &= -1/t^2 \quad (1)
 \end{aligned}$$

$$\therefore \text{eq } y - \frac{2}{t} = -\frac{1}{t^2}(x-2t) \quad (1)$$

$$t^2y - 2t = -x + 2t$$

$$\therefore x + t^2y = 4t \quad \text{QED}$$

$$ii. \quad x \text{-int } y=0 \quad x=4t \quad \therefore Q(4t, 0)$$

$$M_{\perp} = t^2 \quad (1)$$

$$\begin{aligned}
 \text{eq: } y-0 &= t^2(x-4t) \\
 y &= t^2x - 4t^3
 \end{aligned}$$

$$\therefore t^2x - y = 4t^3$$

$$\begin{aligned}
 iii. \quad \text{Solve } t^2x - y &= 4t^3 \text{ and} \\
 xy &= 4 \text{ simultaneously}
 \end{aligned}$$

$$y = t^2x - 4t^3 \quad \& \quad xy = 4$$

$$\therefore x(t^2x - 4t^3) = 4$$

$$t^2x^2 - 4t^3x - 4 = 0 \quad (1)$$

as the 2 x values for this give R & S let them be α & β

$$\alpha + \beta = \frac{4t^3}{t^2} = 4t$$

$$\therefore \text{midpt } \frac{\alpha + \beta}{2} = 2t \quad (1)$$

$$\text{and } y \text{ value} = -2t^3$$

iv. Locus of m

$$x = 2t \rightarrow \frac{x}{2} = t$$

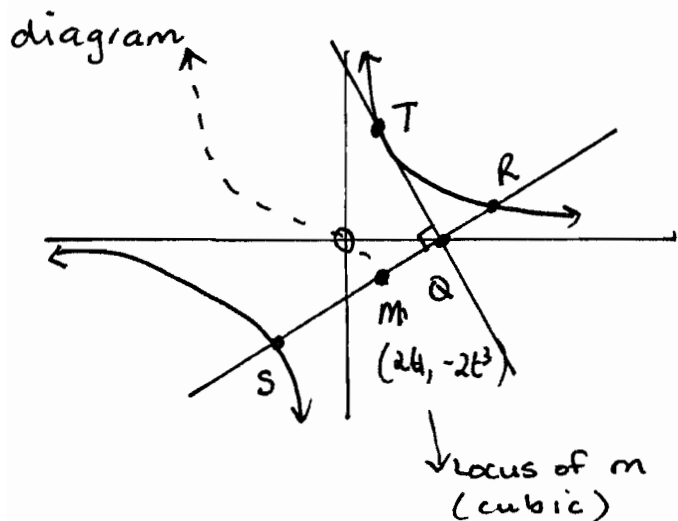
$$\text{Sub into } y = -2t^3$$

$$y = -2 \left(\frac{x}{2} \right)^3$$

$$y = -2 \cdot \frac{x^3}{8}$$

$$y = \frac{x^3}{-4}$$

$$-4y = x^3 \quad \textcircled{1}$$



Locus of m moves on the cubic equation in the 2nd & 4th quadrants ①

Restriction → excluding the origin as $t \neq 0$. ①

Alternative Solⁿ to Q3 (c) iii

$$t^2 x^2 - 4t^3 x - 4 = 0$$

$$x = \frac{4t^3 \pm 4t\sqrt{t^4 + 1}}{2t^2}$$

$$= 2t \pm \frac{2\sqrt{t^4 + 1}}{t}$$

$$R \left[2t + \frac{2\sqrt{t^4 + 1}}{t}, -2t^3 + 2t\sqrt{t^4 + 1} \right]$$

$$S \left[2t - \frac{2\sqrt{t^4 + 1}}{t}, -2t^3 - 2t\sqrt{t^4 + 1} \right]$$

$$\text{Midpt } M = \left[\frac{4t}{2}, -\frac{4t^3}{2} \right]$$

$$= (2t, -2t^3)$$